

anchoring effects.⁴ Most of our treatment effect analysis pools data from the different designs whereas our model estimates of belief updating only uses data where a radiologist reads the same case both with and without AI assistance.

We find that AI assistance does not improve humans’ diagnostic quality on average even though the AI predictions are more accurate than almost two-thirds of the participants in our experiment. Moreover, the zero average effect cannot be explained by the participants ignoring these predictions – we observe that radiologists’ reported probabilities move significantly towards the AI’s predictions with AI assistance. Instead, the zero effect of AI assistance is driven by heterogeneous treatment effects – diagnostic quality increases when the AI is confident (e.g. the predicted probability is close to zero or one) but decreases when the AI is uncertain. In parallel, AI assistance improves diagnostic quality for patient cases in which our participants are uncertain, but decreases quality for patient cases in which our participants are certain. In contrast, providing clinical history does improve diagnostic quality suggesting that humans have additional valuable information that has not yet been incorporated into AI predictions.

An upshot of the results is that information available only to radiologists is useful, but human experts do not correctly combine their information with AI predictions. Specifically, the result that AI assistance can reduce predictive performance cannot be rationalized if our participants are Bayesians with correct beliefs because the AI assistance provides weakly more information to the decision-maker.

Motivated by these results, we analyze two types of deviations from the benchmark model with correct updating to link errors in probabilistic judgement and optimal deployment of AI assistance.⁵ The first type of deviation occurs when agents do not put correct weights on their own information and AI information. We describe this deviation using the approach introduced in Grether (1980; 1992) (see Benjamin, 2019, for a review) to define biases in belief updating. We say that an agent exhibits own-information bias if they over-weights their baseline information when provided with AI assistance and own-information neglect if they under-weights it. Analogously, an agent exhibits automation bias if they over-weights the AI information relative to their own and automation neglect if they under-weights it. The second type of deviation occurs if agents utilize an incorrect joint distribution of their

⁴To ensure that our results do not rely on the wash-out being successful, a third design obtains an assessment with AI assistance only after assessments without AI assistance have been obtained. However, this third treatment is subject to order effects. We find no evidence of order effects on diagnostic quality although there is evidence that familiarity with the interface increases the speed with which participants go through patient cases.

⁵We will remain agnostic about whether the deviations we consider are due to non-Bayesian updating or can be explained by Bayesian updating with incorrect prior beliefs.

own information and AI information – an example of such a deviation is correlation neglect (Enke and Zimmermann, 2019). Our theoretical analysis shows that if agents exhibit only automation neglect, then AI assistance unambiguously increases diagnostic quality. All other forms of biases we consider result in AI assistance reducing diagnostic quality for certain realizations of AI and own information.

We then use the data from our experiment to estimate empirical analogs of the deviations described above and select the model that best describes the treatment effects we document. This exercise requires us to solve several challenges unique to a naturalistic setting where the experimenter does not control the information structure. We find that in the model that best describes our data, agents exhibit automation neglect and act as if their own information and AI predictions are independent (conditional on the truth), even though this is not the case. Interestingly, we do not find evidence of substantial own-information bias or neglect.

Finally, we take our data to estimate the trade-off between diagnostic quality and radiologist time when AI predictions can be selectively utilized, and design an optimal human-AI collaborative system. For each pathology and conditional on the AI prediction for that pathology, our experiment allows us to compute both the diagnostic quality and time taken if a patient case is diagnosed by the AI, a randomly chosen radiologist without AI assistance, and a radiologist assisted with the AI prediction. We use these data to train a classification algorithm that yields the frontier of diagnostic quality and total radiologist time. The results from this exercise mirror our treatment effect analysis – because radiologists take more time with AI and do not correctly incorporate the AI’s information, the majority of cases are optimally decided either by the radiologist or the AI alone but not by the radiologist with access to AI.

Related Literature

A growing body of literature in computer science has explored the predictive performance of humans versus machine learning algorithms, with radiology often serving as a key area of application (Rajpurkar et al., 2018, 2017). Additionally, the study of human-AI collaboration has become an increasingly important facet of medical AI research (Tschandl et al., 2020; Reverberi et al., 2022). For comprehensive overviews of these areas, see Rajpurkar et al. (2022); Hosny et al. (2018); Zhou et al. (2021); Lai et al. (2021). Research on the effectiveness of human-AI collaboration is evolving, with notable studies in radiology including Rajpurkar et al. (2020); Kim et al. (2020); Park et al. (2019); Seah et al. (2021); Fogliato et al. (2022). An active literature studies whether AI assistance benefits radiologists, and which radiologists benefit the most (Rajpurkar et al., 2020; Seah et al., 2021; Ahn et al., 2022; Sim et al., 2020; Gaube et al., 2023). In contrast to prior studies, we recruit a large group of high-

skilled experts from teleradiology companies under contracts that allow us to incentivize our participants. A key differentiating factor of our research is that, unlike previous studies which mainly concentrated on performance, our work emphasizes behavioral biases and their impact on human-AI interaction.

An emerging literature in economics also compares human and AI performance. Within economics, these studies tend to rely on observational approaches, with examples addressing issues in medicine (Ribers and Ullrich, 2022; Mullainathan and Obermeyer, 2019) and bail decisions (Kleinberg et al., 2015; Angelova et al., 2022). However, analyses based on observational data face critical identification challenges, such as the selective labels problem (see Kleinberg et al., 2017; Mullainathan and Obermeyer, 2019; Rambachan, 2021)). A limited set of studies use quasi-experimental approaches (e.g., Stevenson and Doleac, 2019; Angelova et al., 2022) or randomized controlled trials (e.g., Imai et al. (2020); Bundorf et al. (2020); Noy and Zhang (2023); Grimon et al. (2022)) to investigate human use of AI tools, typically focusing on overall performance or variability in participant response. We add to this literature by developing an experimental approach that manipulates the information environment that calculates and compares behavior with a Bayesian benchmark to document systematic biases and demonstrate that these biases lead to a non-trivial delegation problem.⁶

While several studies in behavioral economics have documented errors in probabilistic judgment and belief formation, they do not focus on the issues surrounding human-AI interaction (c.f. Tversky and Kahneman, 1974; Benjamin et al., 2019; Enke and Zimmermann, 2019; Conlon et al., 2022, for example). Our definitions of own-information and automation bias build on the framework in Grether (1980). We contribute to this literature in two ways. First, we develop an approach to estimate the parameters of the model in Grether (1992) in an environment where the joint distribution of the signals cannot be controlled (or partialled out) by the researcher.⁷ This approach is necessary because we cannot modify the signal within medical images. Second, we link the design of AI information provision to the (biased) updating rule that humans use. This link shows that the use of AI information by humans is an important and practical application of the ideas in this literature.

⁶Our finding that radiologists exhibit automation neglect is related to those in Dietvorst et al. (2015), which shows that humans are averse to following algorithmic recommendations as compared to human recommendations. This aversion can be reduced if humans are allowed to modify the algorithm’s recommendation (Dietvorst et al., 2018).

⁷Most applications that we are aware of rely on one of two experimental approaches. In the first approach, the researcher can partial out either the prior information or the likelihood ratio of the signal provided, for example in the classic bookbag-and-poker-chip experiments (see Benjamin et al. 2019; Benjamin 2019, for reviews). In the second approach, the researcher directly provides signals from a known joint distribution (e.g. Conlon et al., 2022).

Our analysis of optimal human-AI collaboration is related to papers that build delegation algorithms to predict the types of cases for which human performance exceeds machine performance (e.g. [Mozannar and Sontag, 2020](#); [Raghu et al., 2019](#); [Bansal et al., 2021](#)). Relative to this work, our analysis uses a decision-theoretic model and specific human biases to trace their consequences for optimal AI deployment.

Finally, our work also adds to the literature on decision-making in the health care context (e.g. [Abaluck et al., 2016](#); [Currie and MacLeod, 2017](#); [Gruber et al., 2021](#); [Chan et al., 2022](#); [Chandra and Staiger, 2020](#)). This work aims to understand predictions and payoffs from observational data on medical decisions, objectives that are achievable under less stringent functional form restrictions in our experimental approach. An important distinguishing feature is that none of these papers consider the effects of AI predictions on medical decision-making.

Overview

The rest of the paper is organized in the following way. Section 2 introduces our model of a decision-maker in a diagnostic setting. Section 3 describes the necessary details of the setting and our experimental design. Section 4 discusses the treatment effects. Section 5 estimates a descriptive model of deviations from Bayesian updating. Section 6 shows the gains achievable under the optimal collaboration between radiologists and AI.

2 Conceptual Model

Our study focuses on classification problems and, specifically machine learning algorithms within the domain of AI tools. These algorithms are designed to predict the appropriate classification for a given case and may assist a human decision-maker. This decision-maker, indexed by r , must take a binary action $a_{ir} \in \{0, 1\}$ on case i based on a prediction of a binary state $\omega_i \in \{0, 1\}$. The realized payoff $u_r(a_{ir}, \omega_i)$ from an action depends both on the state and the action. The expert does not know ω_i but observes, depending on the information environment, a subset of two signals that are potentially informative about the state. The first signal is generated by a prediction algorithm (AI), with realizations $s_i^A \in S^A$. The second signal is directly obtained by the expert, with a realization $s_{ir}^E \in S^E$. These signals are of arbitrary dimension. The joint distributions of the signals conditional on the state is given by $\pi_r(\cdot|\omega) \in \Delta(S^A, S^E)$, with prior probabilities over the state $\pi(\omega)$. We do not place any restrictions on $\pi_r(\cdot|\omega)$. The signals need not be independent conditional on the state of the world, and the signal distribution may depend on the expert, capturing skill ([Chan et al., 2022](#)). We also allow for cases when one of the signals is more informative than